Failures Analysis of Wind Turbines: Case Study of a Chinese Wind Farm

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Abstract—**In the last decade, wind energy industry in China has experienced a rapid growth. Up to now, few studies on reliability analysis of Chinese wind farms can be found. In this study, based on the field data of a wind farm in Jiangsu Province of China, the failure and maintenance statistics are classified, and the reliability of main components in the wind turbines is also analyzed. In addition, the effect of weather factors on the reliability of wind turbines, such as wind speed and temperature, is also demonstrated in a qualitative way. The results show that there are major differences in the performance of wind turbines between Chinese homemade and those of foreign countries. It provides good basis for the operating and maintenance of Chinese wind farms.**

Keywords-wind turbines; failure analysis; reliability; weather; location

I. INTRODUCTION

In the past decades, global warming, energy shortage and environmental pollution have become more serious all over the world, and China is no exception. The development and utilization of renewable energy provides an important means to deal with the problems. Wind energy is one of the most promising renewable energy sources due to its extensive availability and low cost, and also due to the advances in technology and the increase in efficiency [1]. It is clear that wind power has played a critical role in the sustainable development of China [2].

Thanks to the booming growth of Chinese wind energy industry, many scholars, at home or abroad, begin to pay close attention to wind power industry, and numerous papers have been published in recent years. The majority of existed studies emphasized the followed issues, including government policy on wind energy industry [3-5], assessment of wind resources [4], the development route and current status of wind energy development [5, 6], wind turbine (WT) manufacturing industry [7], the capability of grid infrastructure [8], health monitoring techniques of WT structures [9], etc.

As we know that failure and reliability analysis for wind turbines are critical for both of the manufacturers and the operators. Highly stochastic and severe operating environment makes them face big challenges about how to enhance the reliability so as to reduce the operation & maintenance cost. Reliability data analysis can help to reveal the performance of WTs, detect the failure modes and improve the design, schedule inspection and maintenance activities, etc.

In recent years, multiple studies have been conducted to analyze the failure and reliability of WTs. Reference [10] analyzed failure of WTs in Sweden, Finland and Germany. Spinato, et al. [11] investigated the reliability of onshore WTs and their subassemblies in Denmark and Germany over an eleven-year period. In [12], an analysis on the performance, failure and reliability was conducted for a wind farm in South India. Pérez, et al. [13] focused on WT reliability by bringing together and comparing the data collected from the literature. It is found that problems with blades and gearboxes lead to the greatest downtimes. Furthermore, new, larger wind turbines tend to fail more frequently than smaller ones.

To our knowledge, by now there are no studies on failure and reliability data analysis for wind farms in China. This study aims to fill the gap by analyzing failure data of a wind farm in Jiangsu Province of East China. Main concerns of this paper including: (1) Examining reliability index of domestic WTs, including failure frequency, average downtime per failure. (2) Investigating the correlation between components' failure rate and weather factors, including wind speed and temperature, in a qualitative way. (3) Comparing the reliability performance of Chinese domestic WTs with foreign ones.

The rest of this paper is organized as follows. Section Ⅱ introduces the structure and configuration of the under-studied WTs, the common failure modes and their root causes, and maintenance techniques for WTs are also summarized. Section Ⅲ carries out failure analysis for WTs, including failure and downtime distribution of different components, average downtime per failure, average failures versus operational month, etc. In Section Ⅳ, the results of the present study are compared with those in previous literatures. Section Ⅴ concludes this paper.

II. FAILURE AND MAINTENANCE ANALYSIS OF WIND TURBINES

A. Structure of the Wind Turbines

This research is supported by the National Nature Science Foundation of China (NSFC) and Open Fund of Jiangsu Wind Power Engineering Technology Center (no. ZK15-03-01)

978-1-5090-2778-1/16/$31.00 ©2016 IEEE

WT is a core subsystem in wind power system. A WT is a complex electromechanical system which is designed to convert wind (kinetic) energy into mechanical energy so as to generate electricity. During the last decades, different configurations of WTs have been put forward, and various innovative technologies have been developed. Now the most common used structural style is the horizontal axis WT with three blades, for which different combinations of rotational speed, power control, drive train configuration and generator can be used. More details on different configurations of wind turbines can be referred to [13].

This study focuses on a wind farm located in Jiangsu Province, China. There are two relatively independent projects in the wind farm. The first-phase project (Project 1) was put into operation from February 2009, which contains 61 WTs manufactured with unit capacity of 1.5 MW by A company. The second-phase project (Project 2), containing 47 WTs with unit installed capacity of 2 MW manufactured by B company, which started to generate power in January 2011. Eleven subsystems in the WTs are considered in this study, including pitch/blades, hub, brake, gearbox, drive train, generator, yaw system, hydraulics, electric system, control system and sensors.

B. Failure Mode and Root Cause Analysis

Based on the subdivision, the potential failure modes of components are identified. In [14], 16 common failure modes for the WTs were considered, and the root causes were also analyzed. In this study, the failure modes and root causes in [14] are referred to, and the wind farm engineers were also consulted. The failure modes include mechanical, electrical and material aspects. Mechanical aspects include rupture, uprooting, fracture, detachment, thermal, blockage and misalignment, etc; electrical aspects include electrical insulation, electrical failure, output inaccuracy, software fault, and intermittent output, etc; and material aspects include material, fatigue, and structural.

The possible root causes are divided into four different categories, i.e. external, structural, electrical, and wear. In which, external causes include high/low wind, environmental shocks, icing, and lightning strike; structural causes include installation defects, manufacturing and material defect, maintenance errors, and mechanical overload; electrical causes include calibration error, connection fault, overload, electrical short, insulation failure, and software failure; and wear causes include aging, corrosion, fatigue, insufficient lubrication, and overheating, etc.

C. Maintenance Techniques for Wind Turbines

WT is a complex multi-component system. Different subsystems suffer different and various kinds of failures. Furthermore, harsh weather, such as storms and gusts, will generate non-stationary loading and drastically reduce the lifetime of the components. Thus, inspection and maintenance activities play important roles in keeping the availability of wind power systems.

Corrective maintenance (CM) and time-based preventive maintenance (TPM) have been widely adopted in wind power industry. In addition, condition-based maintenance (CBM), combined with prognostics and health management (PHM), can effectively obtain the status and operational state of the components [15]. Opportunistic maintenance is another advanced maintenance strategy which can be used for WT systems. If opportunistic maintenance is used, whenever a component is failed, some other components will be considered to be maintained simultaneously according to pre-specified decision conditions, such as certain age thresholds, in order to reduce the maintenance cost [16]. Moreover, in [17], optimal inspection intervals are determined for the subsystems of the WT based on the delay-time maintenance concept. Thus, it is beneficial to arrange the maintenance activities.

The CM and TPM have been used in the under-studied wind farm, where the period of PM actions is 6 to 12 months, depending on the types of components.

III. ANALYSIS OF WIND TURBINES' FAILURE

A. Sources for Statistical Data

The failure data used in this study is extracted from a wind farm in East China. By now, totally there are 108 WTs.

Supervisory control and data acquisition (SCADA) system has been installed in the wind farm. It is used to record the operation and state data of WTs, including temperature and wind speed, vibration of main components, voltage and current, etc, and to send the failure alarms to the operators automatically. Thus, the operation statistics of the wind farm will be automated and periodically collected by SCADA system. However, up to now the raw operation and failure data are just saved as series of tables in the database according to the date. In fact, a large amount of valuable information is hidden in the data.

The amount of installed turbines and total number of failures during the time span are as follows. For Project 1, the time period is from August 2009 to January 2013, the number of reported failures is 8800; For Project 2, the time period is from January 2011 to July 2013, the number of reported failures is 5417. It needs to be pointed that not all failures are reported during the burn-in period of the new turbines, and the failure statistics of Project 1 WTs during February 2009 to August 2009 are highly incomplete. Therefore, the failure statistics during this time period are omitted in this study.

B. Procedure to Restore a Failure

The maintenance activities in this wind farm are outsourced to a third-party professional maintenance team. The procedure for handling a failure is consistent with that in [10], which can be summarized as follows.

SCADA system automatically records the failures, and will prompt the consequence of the failure and corresponding failure type. SCADA system is monitored by the wind farm operators. Once the operators receive a failure alarm, the maintenance team will be required to perform necessary maintenance actions accordingly.

In case of safety hazard or major damages, the turbine will be shut down automatically or manually. Since many alarms are just acted as reminders, in most cases WTs will be restarted even though the monitor system sends alarm back. If the failure is severe, inspections need to be arranged. According to the severity degree, maintenance personnel or even original equipment manufacturer (OEM) will be contacted to repair or replace the damaged parts. After repair, the failure information will be updated in the database.

C. *Failure Frequency*

The wind farm consists of 108 WTs that are made by two different domestic manufacturers, the failure analysis will be performed separately. Number of failures is an intuitive measure of reliability of different components. Figs. 1 and 2 show the distribution of the failures for Projects 1 and 2 during their study period, respectively.

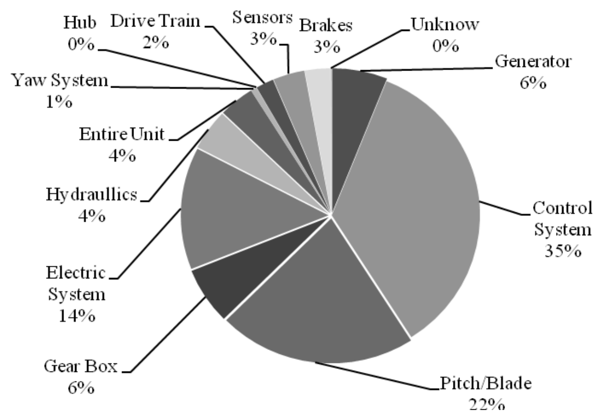

Figure 1. Distrubution of failures for Project 1.

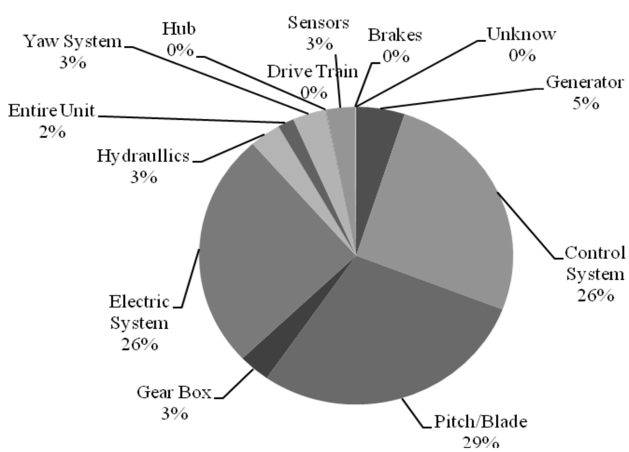

Figure. 2 Distrubution of failures for Project 2.

Fig. 1 shows that most failures in Project 1 arise from control system, pitch/blades and electric system, accounting for about 70% failures of the total. Whereas the result from Project 2 is slightly different, failures of pitch/blades, control and electric systems are common, accounting for about 80% of the totals.

It can be observed that for this wind farm, it experiences more failures due to electrical and electronic components than mechanical components. This result is different from that in [12], where the wind farm experiences more mechanical components failures than electrical and electronic components failures. Based on the statistical analysis of both Figs. 1 and 2, it is found that most failures are caused by control system and electric system, and failures caused by wear on mechanical components are rare. This phenomenon can be explained by the fact that all the 108 WTs are still in their young age, therefore wear-out failures have seldom occurred.

Furthermore, WTs in Project 2 suffer more average failures per turbine.year (i.e. failure rate) than those in Project 1. It seems that larger WTs tend to suffer higher failure rate in comparison with those smaller ones, which is consistent with the findings in [10, 13].

D. *Downtime*

In this section, Pareto analysis is carried out to prioritize the vital component which causes longest downtime. It can help to rank components in descending order based on the downtime, thus it can separate significant components with trivial ones. The percentage of downtime (in hours) per component for Projects 1 and 2 WTs during this study are illustrated as in Figs. 3 and 4, respectively.

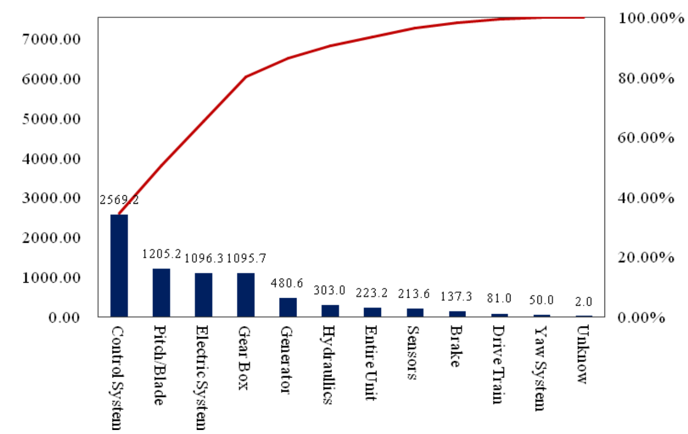

Figure. 3 Distrubution of downtime for Project 1.

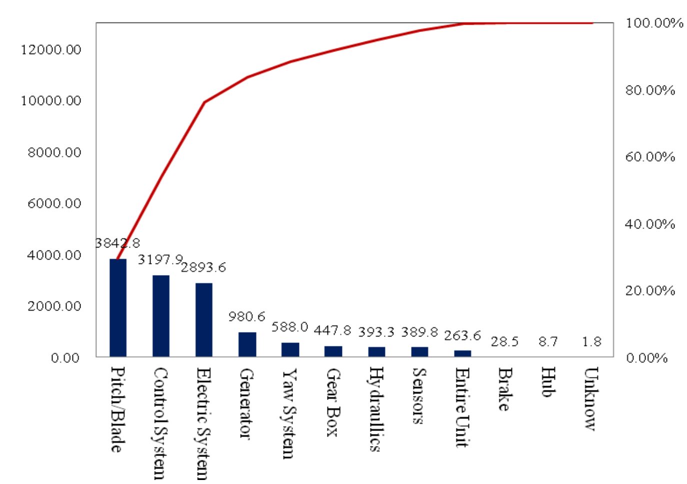

Figure. 4 Distrubution of downtime for Project 2.

The result shows that among the WTs in Project 1, the component with longest downtime is the control system, followed by pitch/blades and electric system, while in Project 2 the component with longest downtime is the pitch/blades, followed by control system and electric system. It's interesting that the rank of the downtime are consistent with the rank of the number of failures for both the two Projects.

It may be not reasonable to draw a conclusion that the most troublesome component is the one with the largest number of failures or with the largest downtime. In practice, we need also consider the severity of the failures. For some components, they may fail frequently, but the maintenance and repair activities are comparatively easy, thus the downtime duration is not so long. On the contrary, some failures occurred to certain components will take a long time to be recovered, so they have great influence on the availability of the wind farm [10]. Therefore, a suitable metric should have the ability to jointly consider the failure frequency and average downtime of a component.

Average downtime per failure is a suitable metric for the failure severity analysis. For WTs of Projects 1 and 2, the average downtime per failure is illustrated in Figs. 5 and 6 respectively.

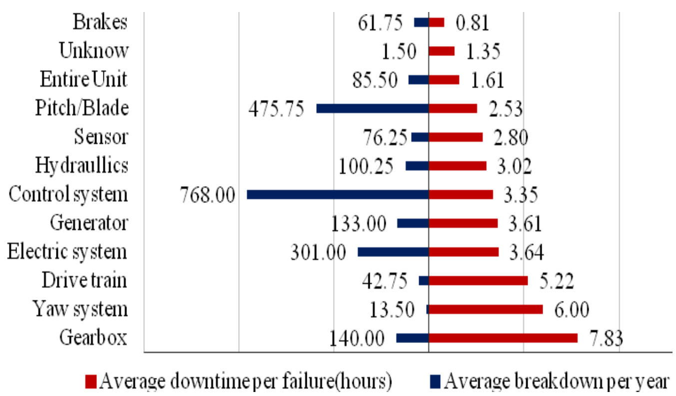

Figure. 5 Average downtime per failure for components in Project 1.

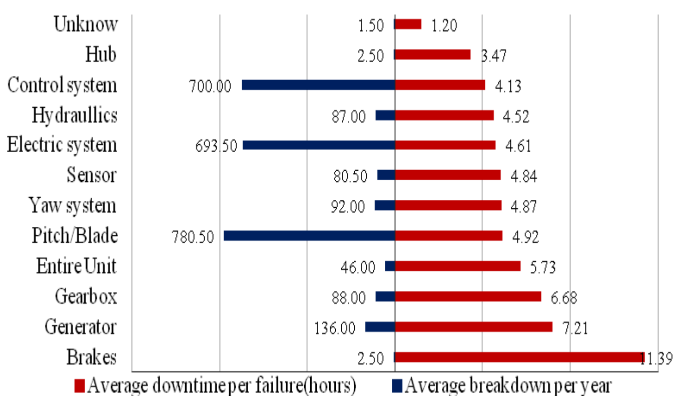

Figure. 6 Average downtime per failure for components in Project 2.

As shown in Figs. 5 and 6, gearbox is the most troublesome component in WTs of Project 1, followed by yaw system and drive train; while the most troublesome component for WTs in Project 2 is brake, followed by generator and gearbox. Moreover, the average downtime per failure for most components of Project 1 lies around 3-6 hours, and only one component, i.e. gearbox, has the average downtime per failure over 7 hours. The result for Project 2 is similar. The average downtime per failure for most components lies around 4-7 hours.

It is worth noting that for WTs of both the two Projects, the most frequently failed components or the components with the longest downtime, i.e. control system, pitch/blades and electric system, the average downtime per failure locates at the middle level among all the components, which is about 3-7 hours.

Furthermore, as a whole the average downtime per failure for components of Project 1 are lower than that of Project 2. It is also shown that the average downtime of mechanical components is relatively larger than those of control and electronic components. It is reasonable in the sense that the time to repair or replace a mechanical component is usually longer than that to recover an electronic component.

E. Failures versus Operational Month

Tavner, et al. [18] concluded that weather and location are important factors that can influence the reliability of WTs. They also demonstrated that there exists a significant cross-correlation between the failure rate and the weather conditions. Their study suggested that the actual correlation is between failure rate and changes in weather, but not only restricted to wind speed. Su, et al. [19] confirmed Tavner et al.'s findings by using time series approach.

In order to analyze the effect of weather factors, including wind speed and environment temperature, on the turbines' reliability, we investigate the relationship between number of failures and the corresponding operational month. The original database contains necessary information for calculating the failures per month for the components. Fig. 7 presents the total number of failures per month for WTs in Project 1. In this study, the wind farm is located in east Jiangsu province, which has a humid subtropical climate. Its main feature is hot, humid in summer and cold, humid in winter.

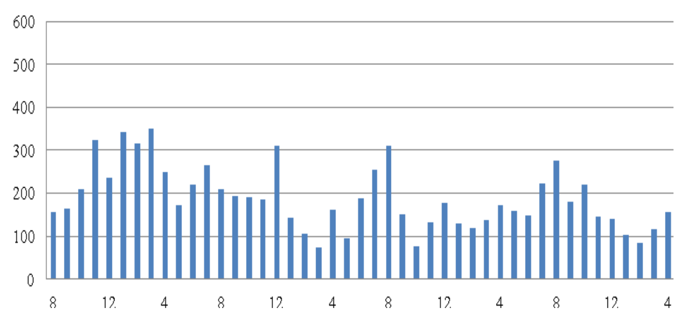

Figure. 7 Failures per month for Project 1 during August 2009 - July 2013.

As can be seen, before 2010 the average failures per month are relatively high, after that the failure numbers seems to be declined slightly. It is worth noting that there are some periodic peaks from the view of failure numbers. For instance, from June to August, and around December. From Fig. 8, we know that during the summer. i.e. from June to August, the temperature is comparably high through a year. Obviously, the average number of failures per month is closely related with the environment temperature. In fact, most rotating components will generate heat when they are working, therefore high environmental temperature will accelerate the process of wear and lead to more failures. In addition, during the winter, i.e., from November to February, it is pretty cold and the wind speed is quite high. Thus it will also lead to more failures due to over-vibration or icing.

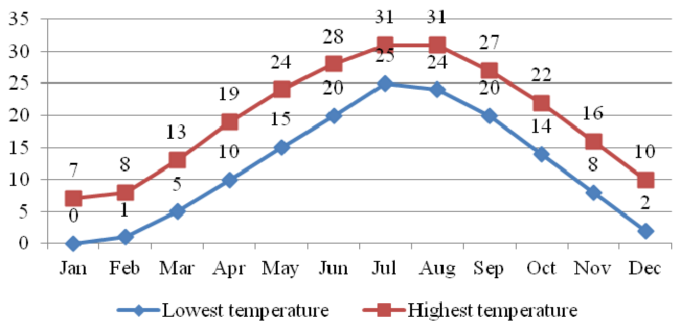

Figure. 8 Local average temperature distribution during a year.

From Fig. 7, we can see that WTs tend to experience more failures during summer and winter. This observation is useful for operators to schedule inspection and PM, and to allocate the spare parts for maintenance, so as to maximize the availability of the wind farm and minimize the total operation and maintenance cost. For example, during summer and winter, PM should be planned more frequently for relevant components, meanwhile the spare parts should also be stored correspondingly.

IV. Comparison with Previous Literature

Up to now, several studies have been conducted to evaluate WTs' reliability in actual operation, including the ones in Germany, Denmark, Sweden, Finland and India [10-13]. In this section, a comparison is made between the present study and the previous studies. The main findings are summarized as follows.

In this study, we find that most failures are related to control and electric systems, pitch/blade, which accounts for over 70% of the totals, the average failure rate for WTs in Project 1 and Project 2 were 37.79 and 57.63 times per year respectively, which are much higher in comparison with existed studies. In those studies, the failure rates usually are less than 3 times a year. Additionally, the result also shows that compared to small turbines, the failure rate is higher for larger turbines.

Based on [13], the largest downtime were found in the gearbox for those of Sweden and Finland, followed by the control system in Sweden and blades/pitch in Finland. In Germany, the largest downtime occurred in the generator and followed by the gearbox. In the present study, the top three largest downtime are found in pitch/blade, control and electric systems respectively.

Reference [10] stated that in Sweden, Finland and Germany, the average downtime for a failure was as high as 62-172h, i.e. two to seven days. However, the result of the present study is less than 13 hours, i.e. half a day. What is the real reason? Taking gearbox as an example, the average downtime per failure for gearbox is about 256h in [10]. As we know that gearbox failures could be due to failures of its sub-components, including bearings, gearwheels, shaft, sealing, oil system and some others. By reviewing the original data, we find that most failures related to gearbox are due to oil system, e.g. oil leak and lack of oil. Usually the downtime caused by failures of oil system is much lower than the others. Comparably, once a time one failure related to shaft even took over 30 days to replace in this wind farm. However, this kind of failures with long downtime has been offset by many others with very short downtime. Therefore, it's not surprised that as a whole the average downtime per failure was much shorter than other studies.

Nevertheless, we can find that according to this study the Chinese home-manufactured WTs are inferior to foreign ones in terms of quality and reliability. For some reasons, WTs in this study are early versions of domestic products, there are still considerable problems related with design, manufacturing and operations management. Thus, they have comparably more failures than those internationally advanced ones. On the other hand, due to the well-designed maintenance system, the availability of the WTs keeps relatively high levels.

In fact, in recent years great efforts have been made by the domestic manufacturers, the reliability of Chinese home-manufactured WTs has been improved greatly. Meanwhile, appropriate condition monitoring and effective maintenance activities are also essential for keeping high availability.

V. Conclusion

In this article, the failure data of WTs in a Chinese wind farm during 2009-2013 are classified and analyzed. An effort is made in the present study to analyze the distribution of failures and downtime for main components of the WTs. Besides, number of failures versus operational month is also analyzed to qualitatively reveal the effect of weather factors on the reliability of WTs. Moreover, the reliability characteristics of domestic-manufactured WTs are compared with those of some other countries.

Some conclusions are drawn from the present study: (1) Pitch/blades, control and electric systems are most relevant to both the failure frequency and downtime of the WTs. (2) In terms of average downtime per failure, the most troublesome components include gearbox, generator and yaw system. (3) Compared with smaller WTs, larger ones tend to suffer higher failure rate. (4) There are statistical correlations between failure rate and weather factors, including temperature and wind speed.

However, there are also some limitations existing in this study. For example, the sample size is not large enough, and the time span is relatively short in comparison to the design life of WTs. Furthermore, it's worth performing failure analysis of WTs installed in different areas and/or manufactured by different companies, thus more valuable information can be obtained. Finally, the correlation between weather factors and WT reliability can be further studied with quantitative approaches, which will provide efficient theoretical basis for preventive maintenance decision of WTs.

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